

ENTERPRISE ARCHITECTURE (EA) AND DATABASE PROGRAMMING

LECTURE 3

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CONTENTS:

- Enterprise Architecture and Other Governance Instruments
- Architecture Methods and Frameworks

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

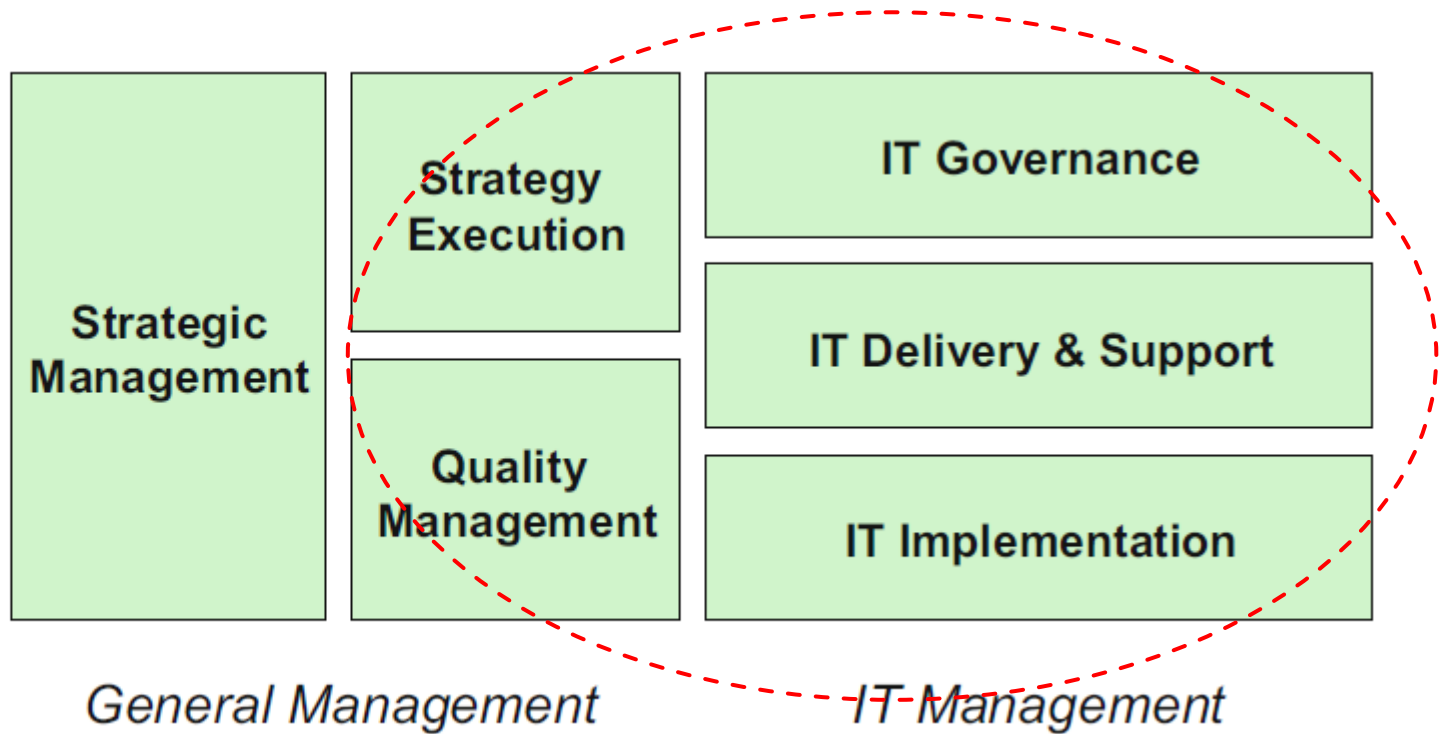
How does EA fit in with other established management practices and instruments?

How does EA position within the context of corporate and IT governance?



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Fig. 2.1 Management areas relevant to enterprise architecture



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

There is a relationship between enterprise architecture with some well-known management practices in each of these areas, which can be shown as the position and role of enterprise architecture in a management context:

- ❑ Strategic management: Balanced Scorecard
- ❑ Business model development: Business Model Canvas
- ❑ Business architecture: BIZBOK® Guide and O-BA
- ❑ **Quality management**: EFQM and ISO 9001
- ❑ **IT governance**: COBIT
- ❑ **IT delivery and support**: ITIL
- ❑ **IT implementation**: CMM and CMMI

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Quality Management

- **EFQM** (European Foundation for Quality Management) as an important management approach.
- This model was first introduced in 1992 as the framework for assessing applications for The European Quality Award.
- The EFQM model has a much broader scope than ISO 9001.
- It not only focuses on quality management but provides an overall management framework for performance excellence of the entire organisation.
- The EFQM model consists of nine criteria for excellence, **five of which are 'enablers'**, covering what an organisation does, and **four are 'results'**, covering what that organisation achieves.

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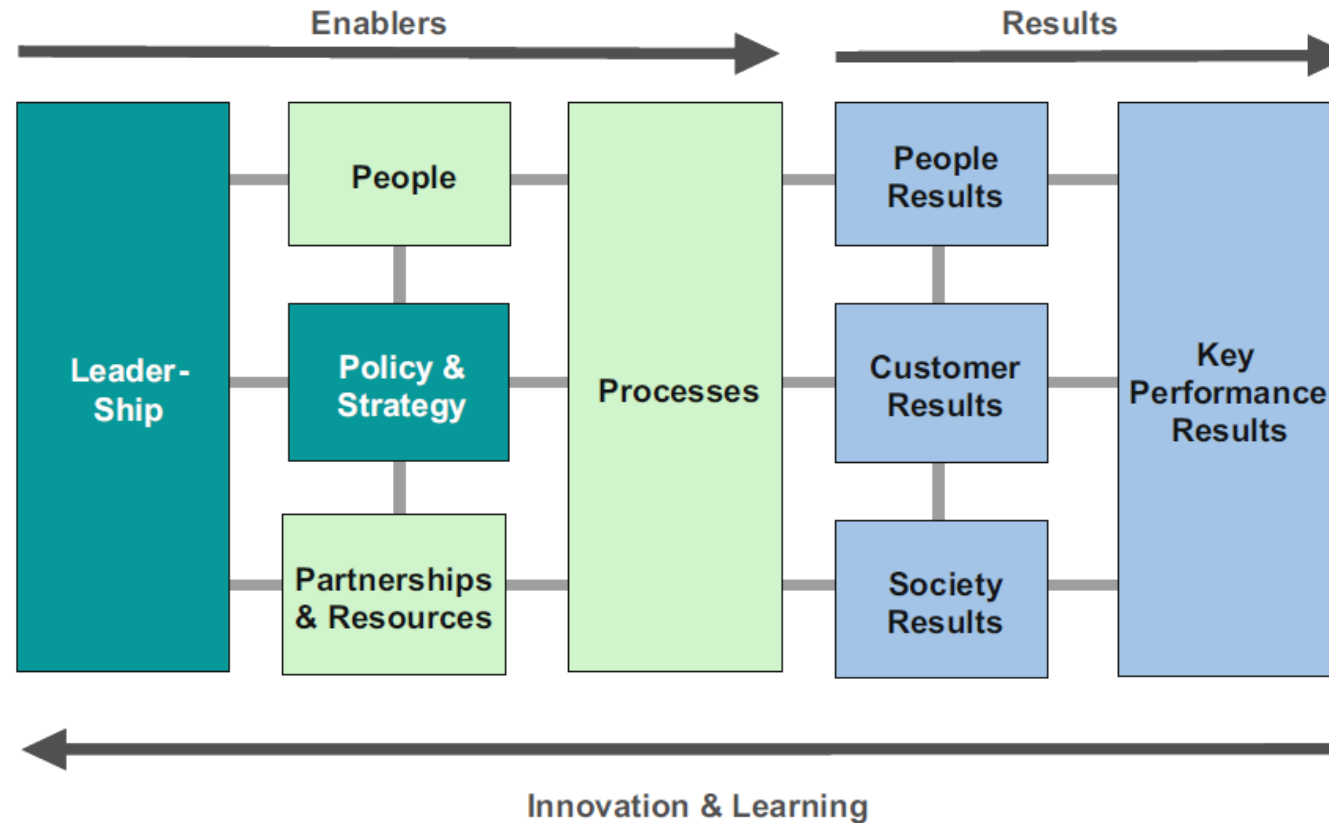
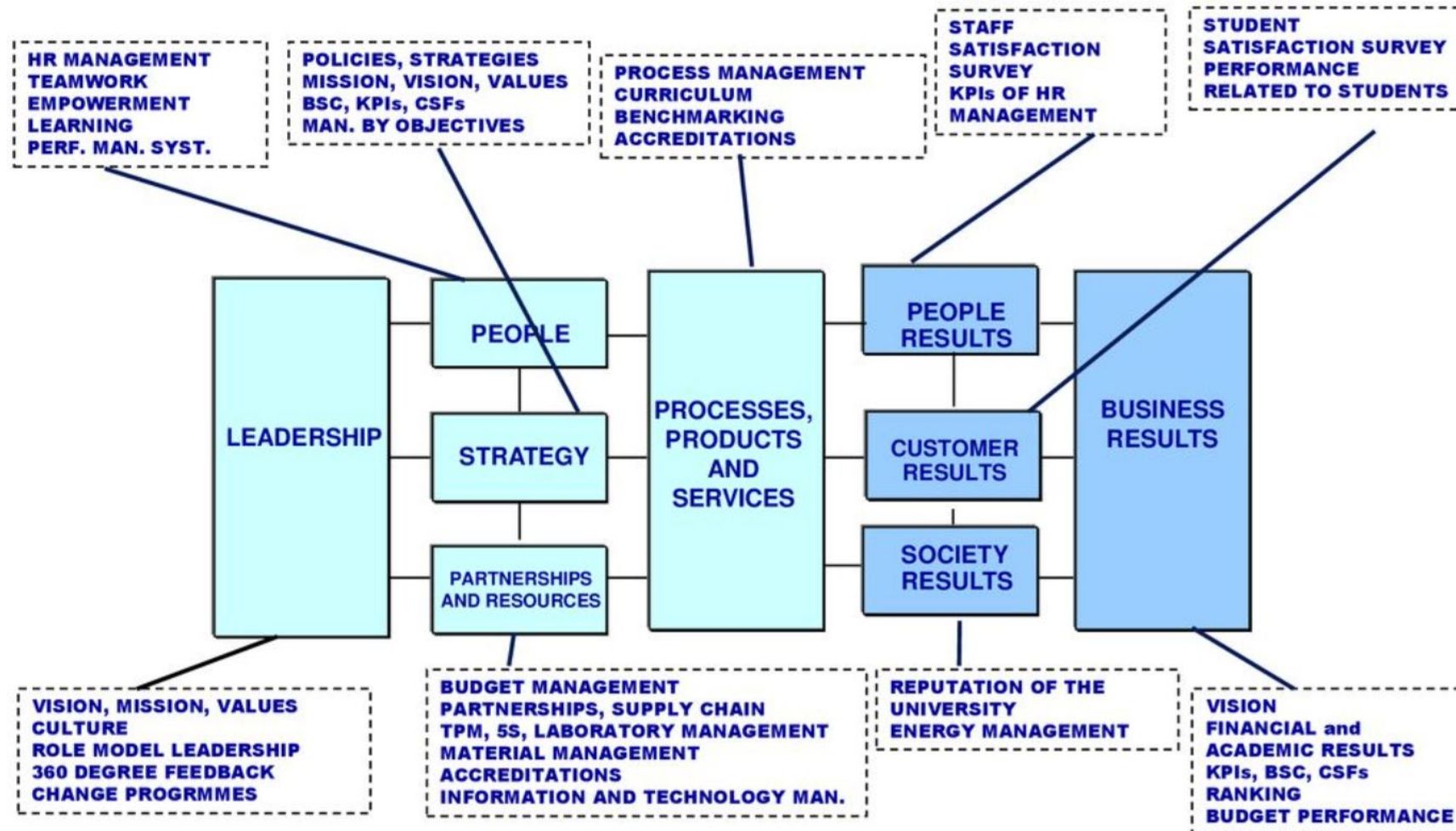


Fig. 2.4 The EFQM excellence model (EFQM 2003)

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ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

Quality Management

- The EFQM model provides **principles, measures, and indicators** for assessing the performance of an enterprise in all these aspects, and these measurements are the basis for **continuous learning, innovation, and improvement**.
- All this also points to the **main difference between the EFQM model and the BSC**: whereas the latter is focused on developing effective strategic management, the former concentrates on measuring and benchmarking the performance of an organisation with respect to a number of best practices. Both are complementary: **the BSC helps to make strategic choices**, and the **EFQM model assists in continuous improvement necessary to execute this strategy**.
- Another important standard in the domain of quality management is the ISO 9001:2000 standard (ISO 2000) of the International Organisation for Standardisation (ISO).

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Discussion

According to the below paper describe

- 1) **What are the differences between ISO 9001 standard and the EFQM model?**
 - 2) **What is the impact of the ISO 9001 standard and the EFQM model on the organisations' performance?**
-
- ❖ Heras-Saizarbitoria, I., Casadesús, M., & Marimón, F. (2011). The impact of ISO 9001 standard and the EFQM model: The view of the assessors. *Total Quality Management*, 22(2), 197-218.



ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

IT Governance

- The **COBIT** (Control Objectives for Information and Related Technology) standard for IT governance was initially published in 1996 by the Information Systems Audit and Control Association.
- COBIT is an internationally accepted IT control framework that provides organisations with 'good practices' that help in implementing an IT governance structure throughout the enterprise.
- It aims to bridge the gaps between business risks, control needs, and technical issues.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

IT Governance

- The core of the COBIT framework is the **control objectives** and **management guidelines** for a set of **IT processes**, which are grouped into five domains:
 - 1) **Evaluate**, direct and monitor
 - 2) **Align**, plan and organise
 - 3) **Build**, acquire and implement
 - 4) **Deliver**, service and support
 - 5) **Monitor**, evaluate and assess

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IT Governance

‘Control’ is defined by COBIT as the policies, procedures, practices, and organisational structures designed to provide reasonable assurance that business objectives will be achieved and that undesired events will be prevented or detected and corrected. The control objectives can help to support IT governance within an enterprise. For example, the control objectives of the ‘Assist and advise IT customers’ process consist of establishing a help desk, registration of the customer queries, customer query escalation, monitoring of clearance, and trend analysis and reporting.

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IT Governance

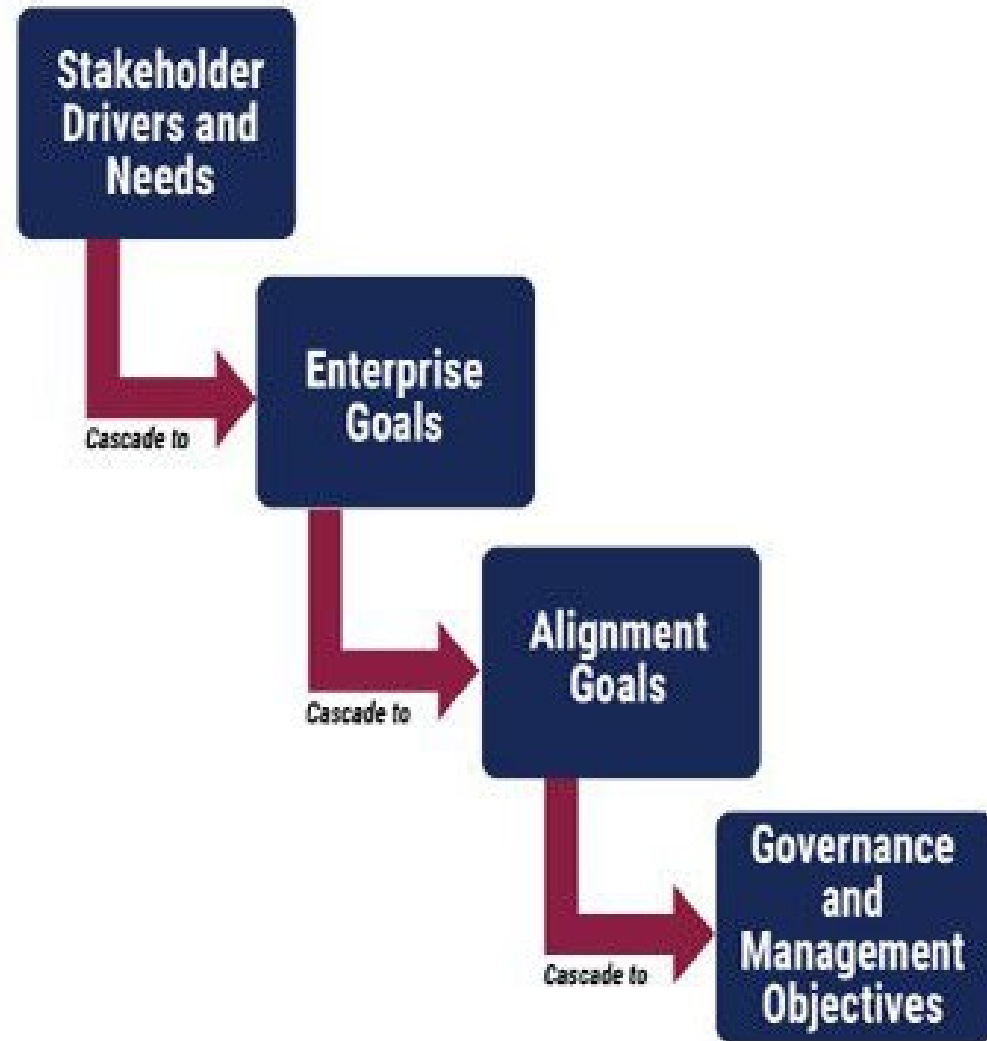
COBIT also offers a maturity model for IT governance, consisting of five maturity levels:

1. **Ad Hoc**: There are no standardised processes. Ad hoc approaches are applied on a case-by-case basis.
2. **Repeatable**: Management is aware of the issues. Performance indicators are being developed, and basic measurements have been identified, as have assessment methods and techniques.
3. **Defined**: The need to act is understood and accepted. Procedures have been standardised, documented and implemented. BSC ideas are being adopted by the organisation.
4. **Managed**: Full understanding of issues on all levels has been reached. Process excellence is built on a formal training curriculum. IT is fully aligned with the business strategy.
5. **Optimised**: Continuous improvement is the defining characteristic. Processes have been refined to the level of external best practices based on the results of continuous improvement with other organisations.

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IT Governance

Goal Cascading in COBIT 2019



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IT Service Delivery and Support

- ITIL (IT Infrastructure Library) (Hanna et al. 2008) is the most widely accepted set of best practices in the IT service delivery domain.
- ITIL comprises a series of documents giving guidance on the provision of good IT services, and on the facilities needed to support IT.
- ITIL has a process-oriented approach to service management.
- It provides codes of practice that help organisations to establish quality management of their IT services and infrastructure, where 'quality' is defined as 'matched to business needs and user requirements as these evolve.'

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IT Service Delivery and Support

- The core of ITIL consists of **two broad groups of processes**:
 - **Service Delivery**, comprising service-level management, availability management, financial management for IT services, IT service contingency management, and capacity management;
 - **Service Support**, covering problem management, incident management, service desk, change management, release management, and configuration management.

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Discussion

According to the below paper (link) describe **what are the differences between ITIL and COBIT?**

- ❖ https://www.simplilearn.com/cobit-vs-til-article#what_are_the_differences_between_iti_and_cobit



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IT Service Delivery and Support

- ITIL is complementary to COBIT. The high-level control objectives of COBIT can be implemented through the use of ITIL.
- COBIT's control objectives tell what to do and ITIL explains how to do it, i.e., what the best practice processes are to realise these objectives.

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IT Implementation

- The **Capability Maturity Model for Software** (Paulk et al. 1993), also known as the **CMM** and **SW-CMM**, is a model for judging the maturity of an organisation's software engineering processes and provides organisations with key practices required to help them increase the maturity of these processes.
- In 2000, the **SW-CMM** was upgraded to **CMMI (Capability Maturity Model Integration)**, which addresses the integration of software development with other engineering activities and expands the scope to encompass the entire product life cycle, including systems engineering, integrated product and process development, and supplier sourcing.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

IT Implementation

In the **CMMI maturity models** in their most common form, there are **five maturity levels**, each a layer in the foundation for ongoing process improvement, designated by the numbers 1–5 (CMMI Product Team 2002):

- 1) **Initial:** Processes are usually ad hoc and chaotic. The organisation does not provide a stable environment. Success in these organisations depends on the competence and heroics of the people in the organisation and not on the use of proven processes.
- 2) **Managed:** The projects of the organisation have ensured that requirements are managed and that processes are planned, performed, measured, and controlled. However, processes may be quite different in each specific instance, e.g., on a particular project.
- 3) **Defined:** Processes are well characterised and understood, and are described in standards, procedures, tools, and methods. These standards are used to establish consistency across the organisation. Projects establish their defined processes by tailoring the organisation's set of standard processes according to tailoring guidelines.
- 4) **Quantitatively Managed:** Quantitative objectives for quality and process performance are established and used as criteria in managing processes. Quantitative objectives are based on the needs of the customer, end users, organisation, and process implementers.

ENTERPRISE ARCHITECTURE AND OTHER GOVERNANCE INSTRUMENTS

IT Implementation

- In any software engineering project of substantial size, **software architecture** plays an important role. The context of this software architecture may be given by an enterprise architecture, which provides **constraints and guidelines for individual software projects**. As such, enterprise architecture is something that becomes especially useful (or even necessary) at CMMI Level 3 and beyond, where projects have to conform to organisation-wide standards and guidelines.

ARCHITECTURE METHODS AND FRAMEWORKS

The IEEE 1471-2000/ISO/IEC 42010 Standard

- In 2000, the IEEE Computer Society approved **IEEE Standard 1471-2000** (IEEE Computer Society 2000), which builds a solid theoretical base for the definition, analysis, and description of system architectures.
- **IEEE 1471**, which has since been subsumed by the **ISO/IEC 42010 standard** (ISO/IEC/IEEE 2011), focuses mainly on software-intensive systems, such as information systems, embedded systems, and composite systems in the context of computing.
- It provides, in the terms of a '**recommended practice**', a number of valuable concepts and terms of reference, which reflect the 'generally accepted trends in practice for architecture description' and which 'codify those elements on which there is consensus'.

ARCHITECTURE METHODS AND FRAMEWORKS

The IEEE 1471-2000/ISO/IEC 42010 Standard

- The standard gives a set of definitions for key terms such as acquirer, architect, architecture description, architectural models, architecture, life cycle model, system, system stakeholder, concerns, mission, context, architectural view, architectural viewpoint.

Fig. 2.5 Conceptual model of architecture description (based on IEEE Computer Society 2000)

ARCHITECTURE METHODS AND FRAMEWORKS

The IEEE 1471-2000/ISO/IEC 42010 Standard

- To explain the role of the stakeholders in the creation and use of an architecture description
- To provide a number of scenarios for the architectural activities during the life cycle: architectures of single systems, iterative architecture for evolutionary systems, architecture for existing systems and architectural evaluation

ARCHITECTURE METHODS AND FRAMEWORKS

The IEEE 1471-2000/ISO/IEC 42010 Standard

Furthermore, the standard gives **six architecture description practices**:

- Architectural documentation referring to identification, version, and overview information.
- Identification of the system stakeholders and of their concerns, established to be relevant to the architecture.
- Selection of architectural viewpoints, containing the specification of each viewpoint that has been selected to organise the representation of the architecture and the reasons for which it was selected.
- Architectural views corresponding to the selected viewpoints.
- Consistency among architectural views.
- Architectural rationale for the selection of the current architecture from a number of considered alternatives.

ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework

- In 1987, John Zachman introduced the first and best-known enterprise architecture framework (Zachman 1987), although back then it was called 'Framework for Information Systems Architecture'.
- The framework as it applies to enterprises is simply a logical structure for classifying and organising the descriptive representations of an enterprise that are significant to the management of the enterprise as well as to the development of the enterprise's systems.
- The framework (Fig. 2.6) in its most simple form depicts the design artefacts that constitute the intersection between the roles in the design process, that is, owner, designer and builder, and the product abstractions, that is, **what (material) it is made of**, **how (process) it works** and **where (geometry) the components are relative to one another**.

ARCHITECTURE METHODS AND FRAMEWORKS

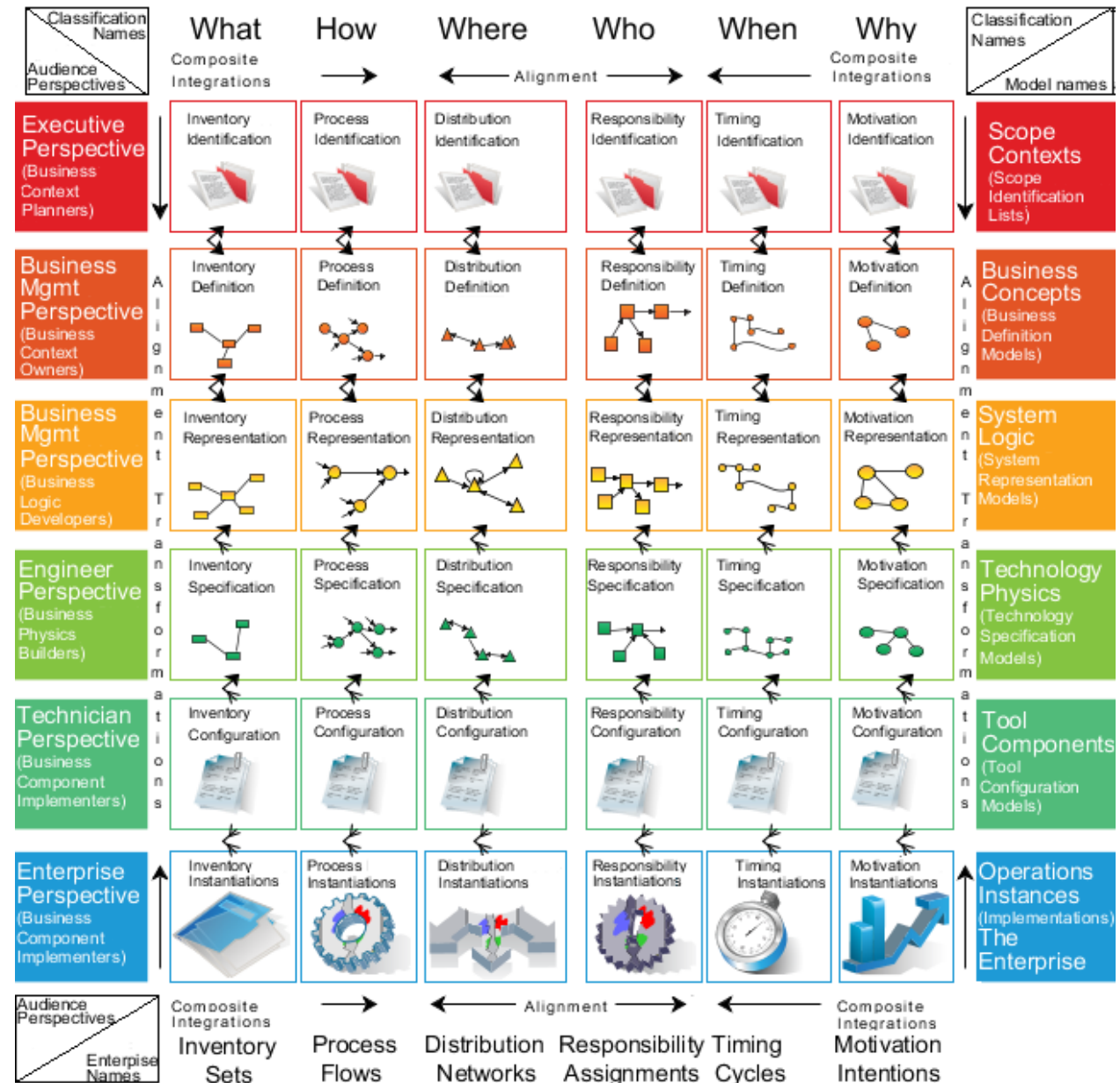
The Zachman Framework

	What?	How?	Where?	Who?	When?	Why?	
Planner							Scope (contextual)
Owner							Enterprise model (conceptual)
Designer							System model (logical)
Builder							Technology model (physical)
Sub- contractor							Detailed representations (out of context)
	Data	Function	Network	People	Time	Motivation	

Fig. 2.6 The Zachman framework (Zachman 1987)

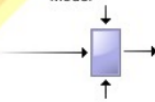
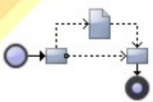
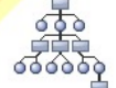
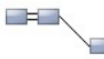
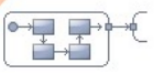
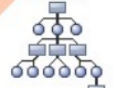
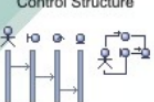
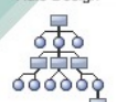
ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework



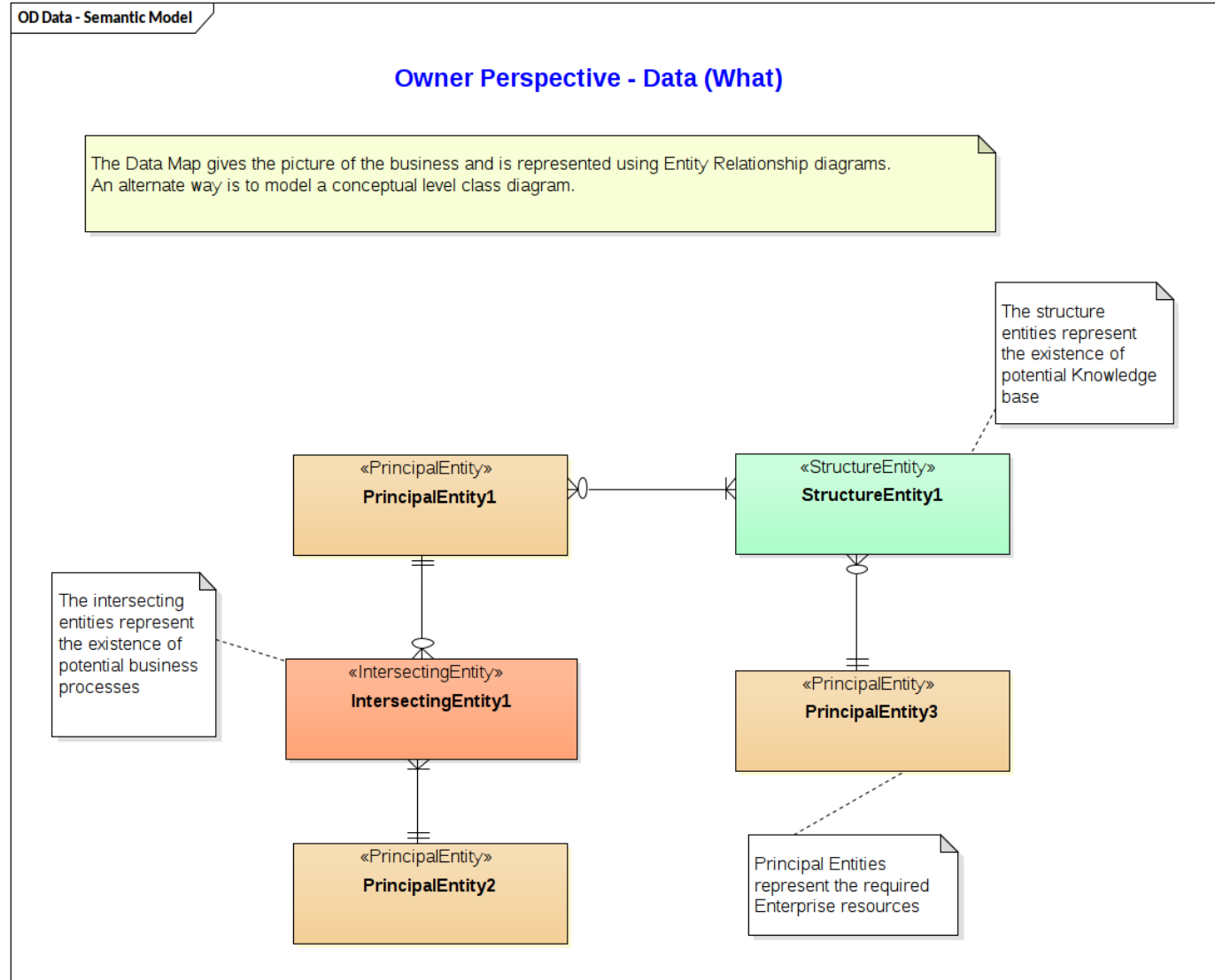
ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework Main View

ZFI Zachman Framework						
The Zachman Framework	DATA What	FUNCTION How	NETWORK Where	PEOPLE Who	TIME When	MOTIVATION Why
SCOPE (Contextual) Planner	Things Important to the Business 	Processes the Business Performs 	Locations in which the Business Operates 	Organizations Important to the Business 	Events/Cycles Significant to the Business 	Business Goals/Strategies 
BUSINESS MODEL (Conceptual) Owner	Conceptual Data Model 	Business Process Model 	Business Logistics 	Work Flow Model 	Master Schedule 	Business Plan 
SYSTEM MODEL (Logical) Designer	Logical Data Model 	Application Architecture 	Distributed System Architecture 	Human Interface Architecture 	Processing Structure 	Business Rule Model 
TECHNOLOGY MODEL (Physical) Builder	Physical Data Model 	System Design 	Technology Architecture 	Presentation Architecture 	Control Structure 	Rule Design 
DETAILED REPRESENTATIONS Sub-Contractor	Data Definition 	Program 	Network Architecture 	Security Architecture 	Timing Definition 	Rule Specification 
FUNCTIONING ENTERPRISE	Data 	Function 	Network 	Organization Units 	Schedule 	Strategy 

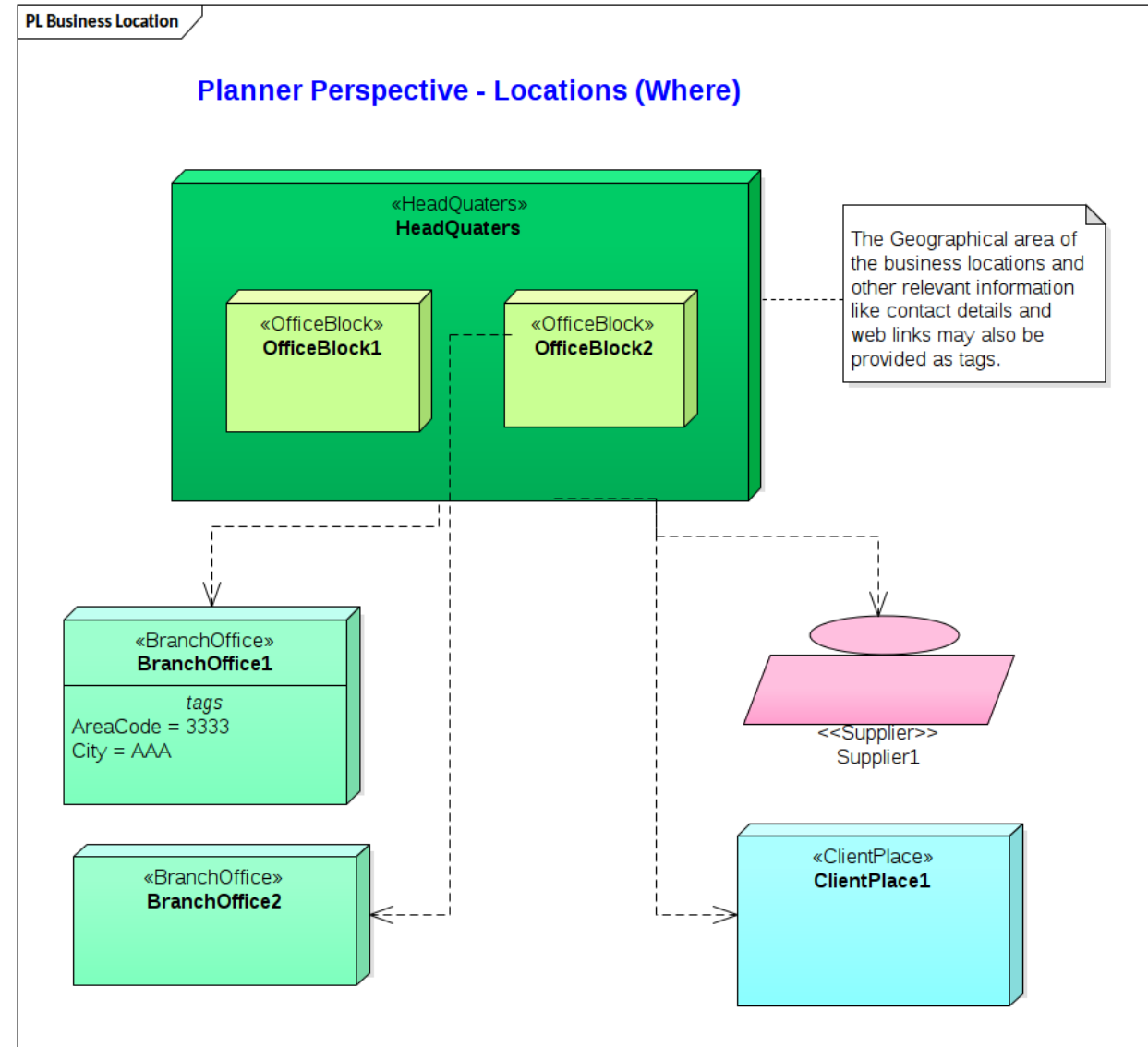
ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework



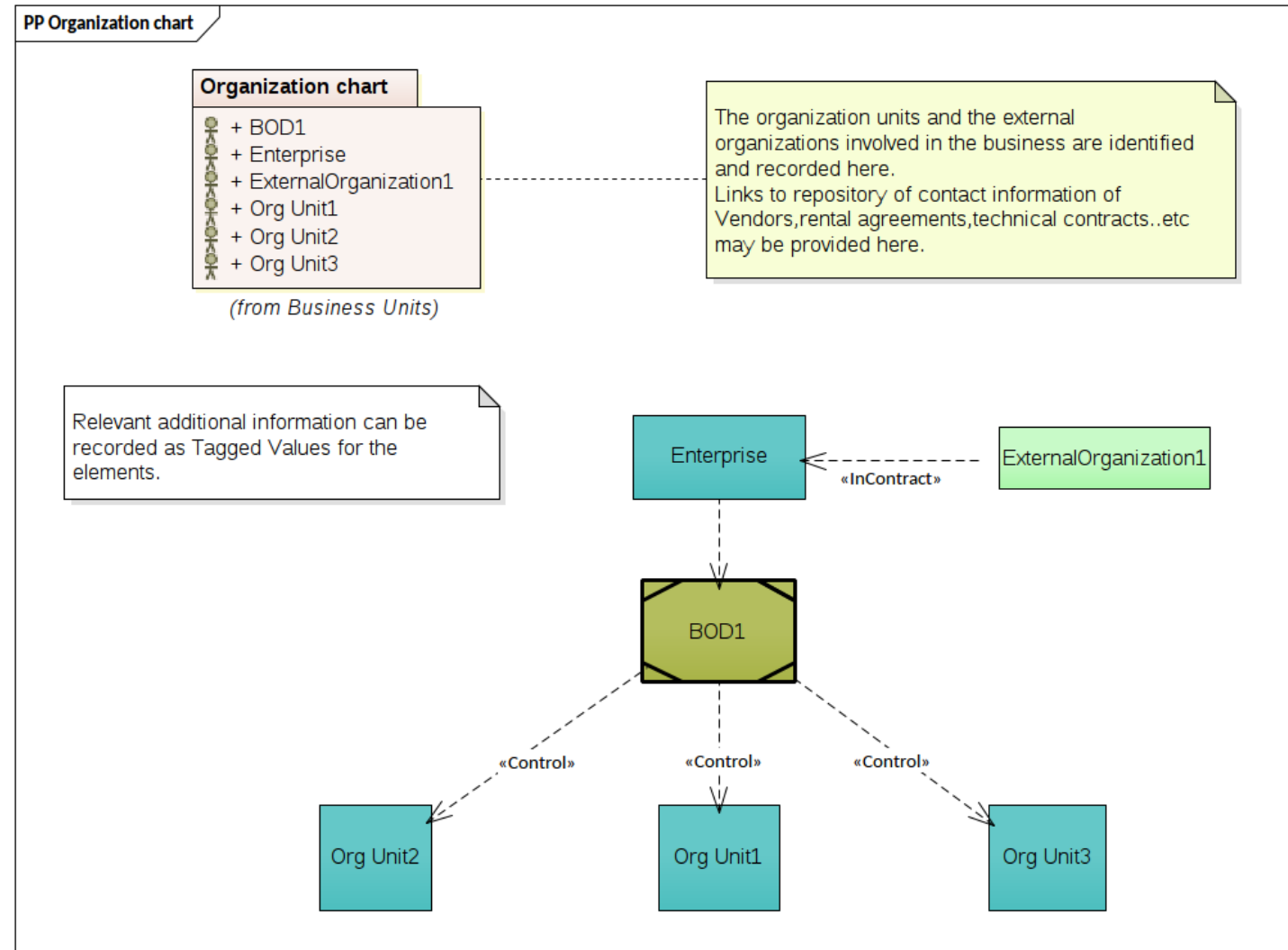
ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework



ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework



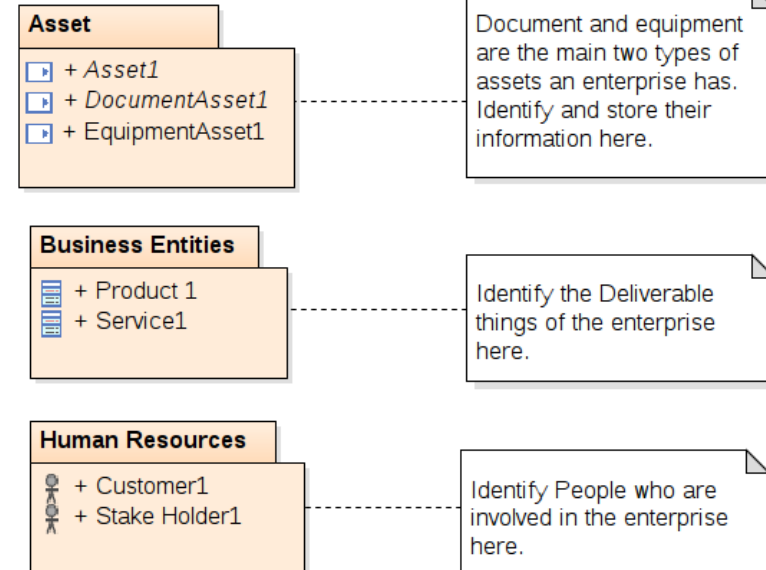
ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework

PD Data - List of Things

Planner Perspective - Data (What)

The things that are important for business are identified in this cell of Zachman Framework. The potential content for this cell includes the resources that are currently available and that are needed by the enterprise for efficient functioning of the business.



ARCHITECTURE METHODS AND FRAMEWORKS

The Zachman Framework

- Advantages of the Zachman framework are that it is easy to understand, it addresses the enterprise as a whole, it is defined independently of tools or methodologies, and any issues can be mapped against it to understand where they fit. An important drawback is the large number of cells, which is an obstacle for the practical applicability of the framework. Also, the relations between the different cells are not that well specified. Notwithstanding these drawbacks, Zachman is to be credited with providing the first comprehensive framework for enterprise architecture, and his work is still widely used.

ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture Framework

- The Open Group Architecture Framework (TOGAF) originated as a generic framework and methodology for development of technical architectures but evolved into an enterprise architecture framework and method.
- From version 8 onwards, TOGAF (The Open Group 2011) is dedicated to enterprise architectures.

ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture Framework

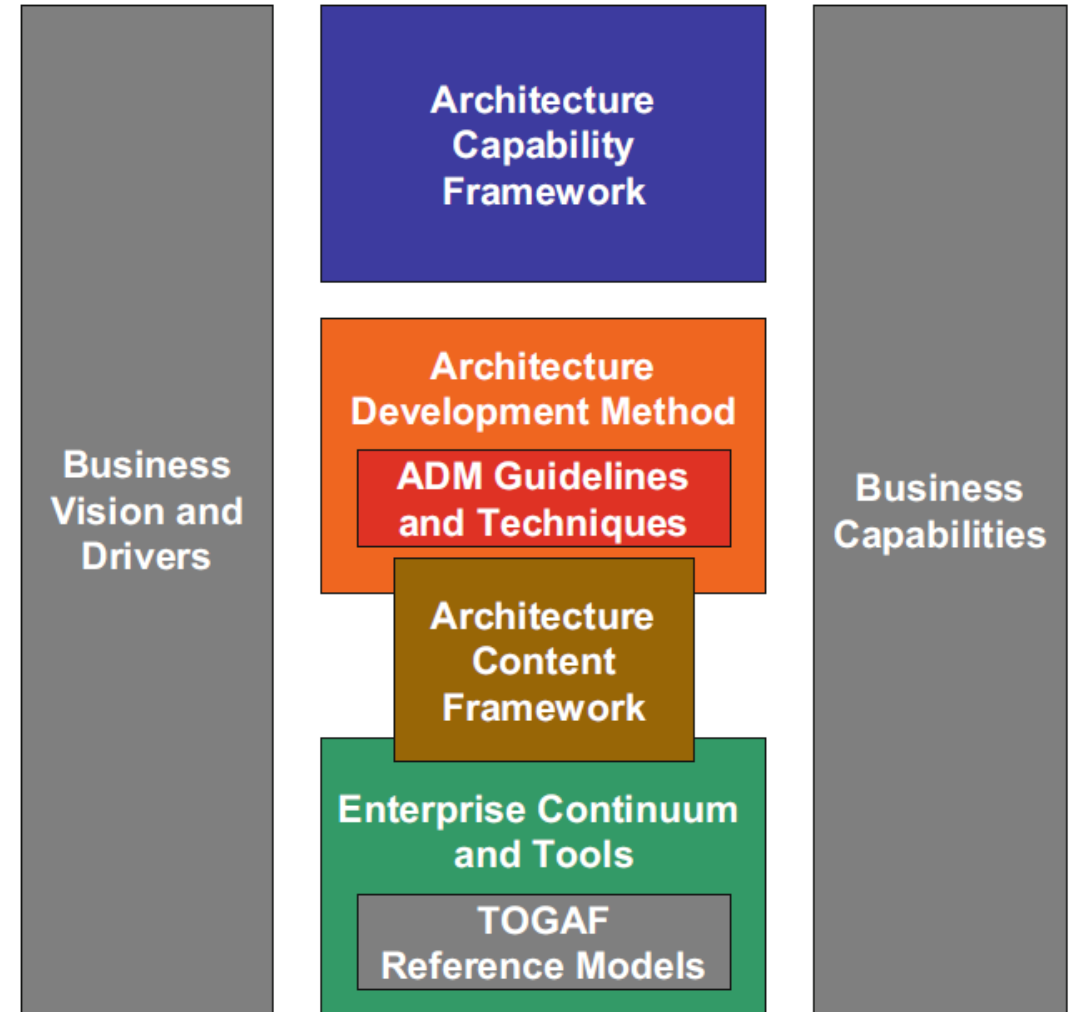
TOGAF has the following main components (Fig. 2.7):

- ❑ **An Architecture Capability Framework**, which addresses the organisation, processes, skills, roles, and responsibilities required to establish and operate an architecture function within an enterprise.
- ❑ **The Architecture Development Method (ADM)**, which provides a 'way of working' for architects. The ADM is considered to be the core of TOGAF, and consists of a stepwise cyclic approach for the development of the overall enterprise architecture.
- ❑ **The Architecture Content Framework**, which considers an overall enterprise architecture as composed of four closely interrelated architectures: Business Architecture, Data Architecture, Application Architecture, and Technology (IT) Architecture.
- ❑ **The Enterprise Continuum**, which comprises various reference models, such as the Technical Reference Model, The Open Group's Standards Information Base (SIB), and The Building Blocks Information Base (BBIB). The idea behind the Enterprise Continuum is to illustrate how architectures are developed across a continuum ranging from foundational architectures, through common systems architectures and industry-specific architectures, to an enterprise's own individual architecture.

ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture
Framework

Fig. 2.7 TOGAF 9.1 (The
Open Group 2011)



ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture Framework

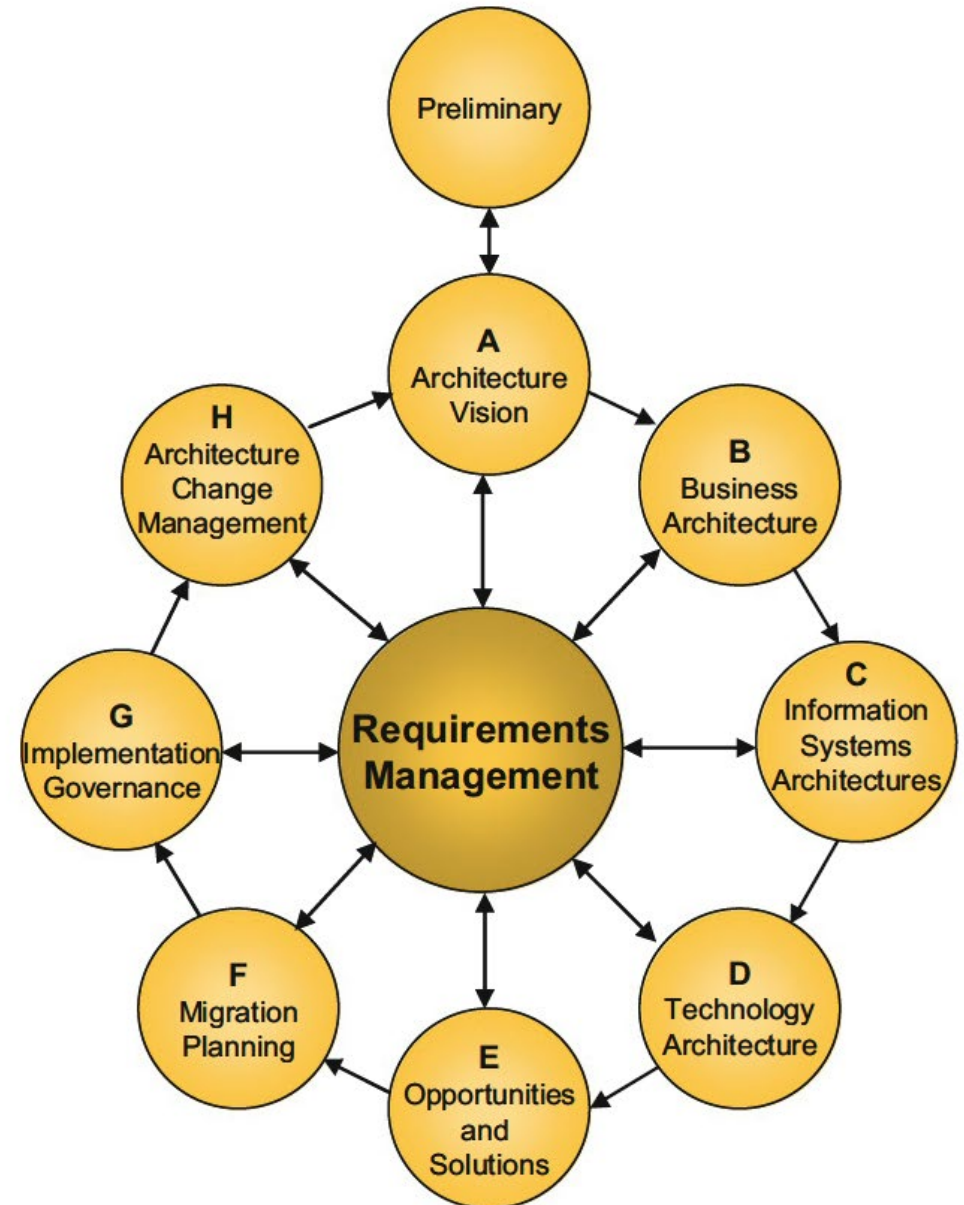
TOGAF's ADM (Fig. 2.8) is iterative, over the whole process, between phases and within phases. For each iteration of the ADM, a fresh decision must be taken as to:

- The breadth of coverage of the enterprise to be defined;
- The level of detail to be defined;
- The extent of the time horizon aimed at, including the number and extent of any intermediate time horizons;
- The architectural assets to be leveraged in the organisation's Enterprise Continuum, including assets created in previous iterations of the ADM cycle within the enterprise and assets available elsewhere in the industry.

ARCHITECTURE METHODS AND FRAMEWORKS

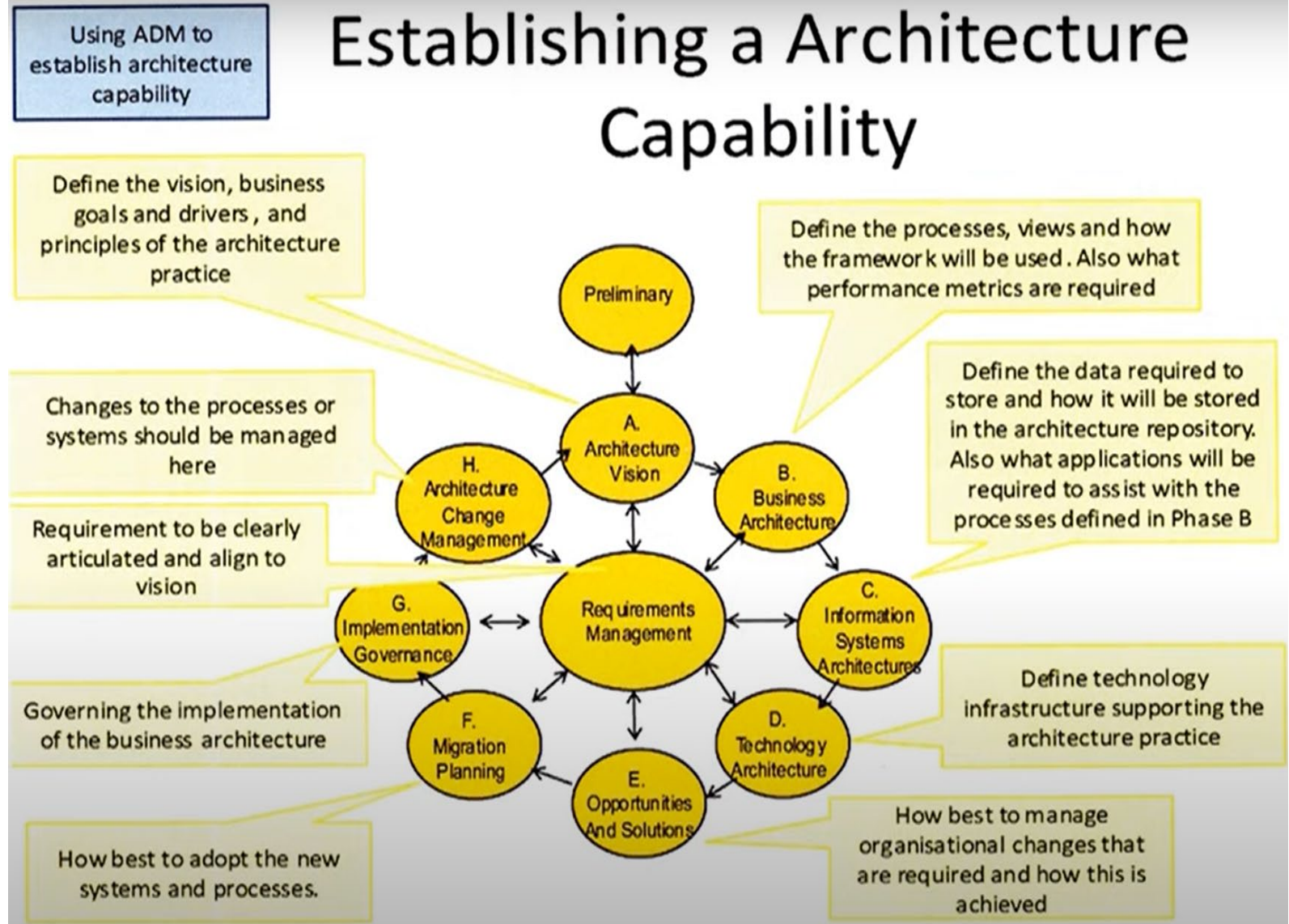
The Open Group Architecture Framework

Fig. 2.8 TOGAF
architecture development
method (The Open Group
2011)



ARCHITECTURE METHODS AND FRAMEWORKS

The Open Group Architecture Framework



ARCHITECTURE METHODS AND FRAMEWORKS

OMG's Model-Driven Architecture

- The **Model-Driven Architecture (MDA)** (Object Management Group 2014; Frankel 2003) aims to provide an open, vendor-neutral approach to interoperability.
- It builds upon the Object Management Group's modelling standards: the Unified Modeling Language (UML; see also Sect. 2.3.3), the **Meta Object Facility (MOF)** (Object Management Group 2015d) and the **Common Warehouse Meta-model (CWM)**. Platform-independent application descriptions built with these standards can be realised using different open or proprietary platforms, such as Java, .NET, XMI/XML and Web services.
- MDA wants to raise the level of abstraction at which software solutions are specified by defining a framework supported by a collection of standards that sets a standard for generating code from models and vice versa. Now, MDA-based software development tools already support the specification of software in UML instead of in a programming language like Java.

ARCHITECTURE METHODS AND FRAMEWORKS

OMG's Model-Driven Architecture

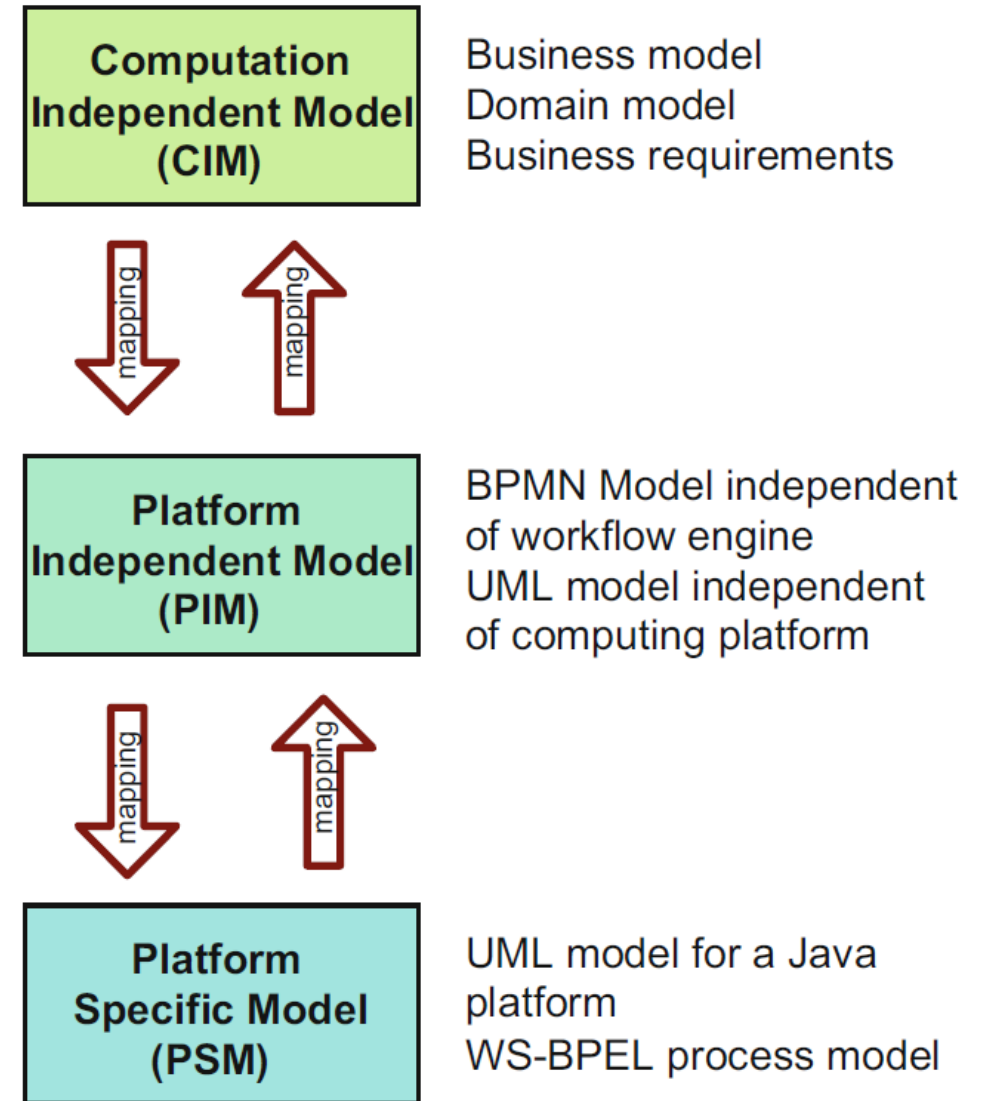
MDA comprises three abstraction levels with mappings between them (see Fig. 2.9):

- 1) The requirements for the system are modelled in a domain model or business model, historically called **Computation-Independent Model (CIM)** in MDA, which describes the situation in which the system will be used. It hides much or all information about the use of automated data processing systems.
- 2) The **Platform-Independent Model (PIM)** describes the operation of a system while hiding the details necessary for a particular platform. A PIM shows that part of the complete specification that does not change from one platform to another.
- 3) A **Platform-Specific Model (PSM)** combines the specifications in the PIM with the details that specify how that system uses a particular type of platform.

ARCHITECTURE METHODS AND FRAMEWORKS

OMG's Model-Driven
Architecture

Fig. 2.9 MDA framework



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